

DJ MUSIC SYSTEM

Portable USB Edition & Event Safety Plan

Project documentation for a two-deck DJ application designed to make emergency laptop replacement quicker during live events.

Project area	Planned design
Main objective	Continue music playback if the main laptop fails.
Portable format	Windows desktop application stored on USB.
Backup approach	Primary USB and a separate emergency USB copy.
DJ controls	Two decks, CUE, LOOP, SYNC, EQ, crossfader and master volume.
Audio connection	Laptop → audio interface/controller → amplifier → DJ speakers.

Core design rule: the software should not depend on one laptop, one drive letter, or one fixed installation folder.

1. Project Overview

The DJ Music System is intended for personal use and live events. The main reliability requirement is portability: if the primary computer has a problem, the operator should be able to connect the USB to another compatible Windows laptop, open the application, select the available audio output and return to the event playlist with minimal delay.

The software will include a DJ-style interface, local music library, playlists, favorites, play history, audio-device settings and an Emergency/Recovery screen.

2. Core Features

Feature	Description
Portable app	Run from USB without a conventional installation being required for normal use.
Drive-letter independence	Discover the USB root at runtime; do not hard-code C:\, D:\ or another drive.
Deck A and Deck B	Load and control two tracks independently.
DJ controls	Play/pause, cue, loop, sync, BPM display and future beat-grid support.
Mixer	Crossfader between A and B plus master output control.
EQ	Bass, mid and treble controls.
Music library	Search, categories, playlists, favorites, artwork and recent history.
Emergency mode	Quick access to the last playlist, recently played songs and audio settings.
Audio settings	Choose laptop output, USB audio interface or compatible DJ controller.

3. USB Folder Structure

The finished portable package should be organized so the application can locate its own files even if Windows assigns a different drive letter.

DJ-MUSIC-USB/ ■■■ DJMusic.exe ■■■ Data/ ■ ■■■ djmusic.db ■ ■■■ playlists/ ■ ■■■ favorites/ ■ ■■■ history/ ■■■ Music/ ■ ■■■ Telugu/ ■ ■■■ Hindi/ ■ ■■■ English/ ■ ■■■ DJ/ ■ ■■■ Band/ ■ ■■■ New-Releases/ ■■■ Artwork/ ■■■ Settings/ ■■■ Logs/

- Keep important event tracks on the USB when capacity allows.
- Store playlists, favorites, settings and history with the portable data.
- Use relative paths or resolve the USB root when the program starts.
- Allow audio output to be selected again when the USB is connected to a different laptop.
- Do not rely on a fixed Windows drive letter.

4. Emergency Recovery Procedure

Step	Action
1	If safe to do so, secure the audio path and prepare the backup laptop.
2	Remove the primary DJ USB from the failed laptop.
3	Connect it to the backup Windows laptop.
4	Open DJMusic.exe from the USB.
5	Select or confirm the audio output device.
6	Open Emergency/Recovery or the required event playlist.
7	Load the next track into Deck A or Deck B and resume playback.

Important: this recovery path must be rehearsed before an event. A portable application alone cannot guarantee that a second laptop will have the right audio drivers or compatible hardware.

5. Two-USB Backup Strategy

Copy	Role	Handling
USB 1	Primary	Used for normal event operation.
USB 2	Emergency backup	Kept separate from the active system and tested before the event.

Update both copies whenever the application, playlists or event music changes. Keep the emergency copy physically separate to reduce the chance that one incident affects both copies.

6. Two-Deck DJ Interface

Screen area	Planned controls and information
Deck A	Track title, artwork, BPM, elapsed/remaining time, playback status, CUE, LOOP, SYNC and transport controls.
Deck B	Same controls as Deck A.
Mixer section	A ↔ B crossfader and master output level.
EQ section	Bass, mid and treble adjustments.
Library panel	Search, category filters, playlists, favorites and recent tracks.
Emergency panel	Last-used event playlist, recently played tracks and quick loading to either deck.

7. Audio and Event Equipment

The intended event setup is a laptop running the DJ application, connected to an amplifier and the DJ sound system. Depending on the equipment, a USB audio interface or DJ controller may sit between the laptop and the amplifier.

DJ Music System | v Laptop / USB Audio Interface / DJ Controller | v Mixer or Amplifier | v DJ Setup / Speakers
Optional headphone CUE: Software → separate headphone output → headphones

Advanced feature: independent master and headphone/CUE outputs require suitable multi-channel audio hardware and a compatible low-latency audio implementation.

8. Music Library and Categories

- Organize local music by language, genre, event and custom category.
- Initial suggested categories: Telugu, Hindi, English, DJ, Band/Pad Band and New Releases.
- Support event-specific playlists, favorites and recently played history.
- Store artwork and track metadata where available.
- Plan for BPM, key and beat-grid metadata as later DJ features.

9. Online Music and New Releases

The app may include music discovery and integrations with services that authorize the intended playback. It should not bypass third-party authentication, DRM, advertisements or service restrictions. For dependable event playback, keep the essential event playlist available locally and ensure music is used under the applicable rights and licensing arrangements.

10. Local Data and Database

SQL Server can be used during development if desired. For the final portable package, an embedded database such as SQLite is a practical choice because it avoids needing a separate database server installed on each backup laptop.

Data group	Purpose
Tracks	Track title, artist, file location, BPM and metadata.
Playlists / PlaylistItems	Named playlists and their track order.
Favorites	Tracks marked for quick access.
PlayHistory	Recently played tracks and timestamps.
Settings	Portable application and audio-device preferences.
Categories	Language, DJ, Band, New Release and user-created categories.

Component	Direction
Portable database	SQLite is recommended for the final USB package.
Audio engine	Low-latency desktop audio design for two simultaneous decks and future CUE routing.
Packaging	Application executable and self-contained USB folder.
Path handling	Discover USB root dynamically; avoid fixed drive letters.

15. Final Objective

The finished application should behave like a dedicated DJ appliance: quick to start, simple to operate, independent of a particular Windows drive letter, configurable for the available event audio setup and prepared for emergency laptop replacement.

Reliability rule: portability can shorten recovery time, but testing the backup laptop, USB copies and full audio chain before every important event remains essential.